Amir Mohseni

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Education

MSc Advanced Computer Science Oct 2026 – Sep 2027
University of Oxford, Oxford, United Kingdom

- Incoming Master's student, with an intended focus on **Artificial Intelligence**.

Bachelor of Computer Science Sep 2023 – Jun 2026
Maastricht University, Maastricht, The Netherlands

- **GPA:** 8.9 / 10.0
- Selected for the **MARBLE Research Program (Honours Track)**.
- Thesis: investigated how users seek legal advice from LLMs, studied overconfidence issues in RL-trained LLMs, and developed low-cost legal-query routing classifiers achieving **91%+ accuracy** at an estimated **1000× lower cost** than repeated frontier-LLM classification.
- Created the **ICPC Club** at Maastricht University; organized weekly training sessions and contests, and qualified for **NWERC** in all eligible years.
- **Key Courses:** Machine Learning, High-Performance Computing, Cloud Computing, Recommender Systems, Databases.

Bachelor of Computer Engineering Sep 2021 – May 2023
Shahid Beheshti University, Tehran, Iran

- **GPA:** 17.8 / 20.0
- Teaching Assistant for **Discrete Mathematics**, **Advanced Programming**, and **Data Structures and Algorithms**.
- Qualified and participated in the **ICPC World Finals**.
- Transferred to Maastricht University to continue the BSc in Computer Science.

Publications

CurveBench: A Benchmark for Exact Topological Reasoning over Nested Jordan Curves 2026

Amirreza Mohseni, Mona Mohammadi, Morteza Saghafian, Naser Talebizadeh Saradari. *arXiv preprint arXiv:2605.14068*. arXiv

Experience

Student Researcher, Turing – Remote Apr 2025 – Present

- Leading a team of student researchers focused on **LLM modeling, reinforcement learning, and evaluation**.
- Fine-tuning in-house models using reinforcement learning to improve **tool-use**, code-generation, and reasoning capabilities.
- Trained **reward models** to predict human preferences in educational tasks using EduArena preference data, improving preference prediction accuracy from 62% to 75%.
- Presented the **EduArena** research platform at ICML 2025.

Generative AI Researcher, Scale AI – Remote Sep 2024 – Apr 2025

- Collaborated with ML researchers and PhD students to fine-tune and benchmark open-source models including **Gemma 2** and **Llama 3**.
- Improved reasoning performance on benchmarks such as **Math-500**, **MMMU**, and **GPQA Diamond** using SFT and DPO methods inspired by **STaR** and **V-STaR**.
- Delivered executive reports to enterprise clients, summarizing experimental results, performance evaluations, and research insights.

Research Assistant, Maastricht University – Maastricht, The Netherlands Jun 2024 – Sep 2024

- Collected and cleaned large-scale social-media data from company Facebook pages using a licensed API.
- Trained and evaluated regression models to analyze post engagement and social trend patterns in response to news events.

Projects

Professional Projects

- **EduArena**: Core contributor to a large-scale research crowdsourcing platform for evaluating LLMs on educational tasks. Implemented automated pipelines for model routing, preference-data collection, and RLVR data generation using entropy and model-disagreement signals.
- **Reasoning Router**: Developed a router that predicts whether a query should be handled with *thinking* or *non-thinking* inference, reducing token usage by nearly **50%** while maintaining performance within **5%** of the best model's accuracy. Blog / Models

Academic Projects

- **Legal Query Routing for User–LLM Interactions**: Built a classifier pipeline for analyzing legal-advice-seeking behavior in real-world chatbot conversations. Developed a legal-topic taxonomy, compared LLM- and encoder-based methods, and investigated overconfidence in RL-trained models for legal classification, proposing a simple mitigation strategy to reduce overconfidence.
- **CurveBench RLVR Fine-Tuning**: Developed RLVR-style fine-tuning experiments for vision-language models on exact topological reasoning tasks, using structured rewards based on rooted-tree recovery from images of nested Jordan curves.

Achievements

- **Ranked 482nd** among ∼120,000 mathematics participants (**Top 0.4%**) in the National University Entrance Exam, Iran (2021).
- **ICPC World Finalist** (45th Annual, 2022) — ICPC Profile.
- **Bronze Medalist**, ACM-ICPC Regional Contest (2023).
- Reached the **Top 120** in the National Olympiad of Informatics, Iran (2020).

Skills

Programming Languages: Python, C/C++, Java

Machine Learning & AI: PyTorch, NumPy, TRL, vLLM, verl, Hugging Face Transformers

Databases & Cloud: SQL, AWS, GCP Vertex AI

Other: Competitive programming, technical writing, research communication, team coordination